Tarea 2: Path Tracing con OpenMP y Numba

> Falta ejecutar este mismo script en un segundo computador fisico y copiar sus resultados a la seccion de comparacion. No se inventan mediciones de hardware no disponible.

Declaracion de uso de IA

Se uso un asistente de IA para revisar la pauta y las clases, implementar las versiones OpenMP/Numba, crear scripts de experimentacion y redactar este informe. Los resultados numericos provienen de ejecuciones locales registradas en `results.csv`.

Resumen del problema

El ray tracing serial calcula iluminacion directa con modelo de Phong, sombras y rebotes especulares. El path tracing agrega rebotes difusos aleatorios en el hemisferio y promedia N caminos por pixel, por lo que el costo crece como O(WHNDK), donde K es el numero de objetos. Cada pixel es independiente: corresponde a un patron map, trivialmente paralelizable en memoria compartida.

La carga por pixel es irregular porque algunos rayos escapan al fondo rapidamente, mientras otros generan cadenas de rebotes, rayos de sombra e intersecciones contra mas objetos. Por eso `schedule(static)` puede dejar threads esperando en la barrera final, mientras `dynamic` o `guided` mejoran el balance a cambio de overhead de planificacion.

Implementacion

`omp_pt.cpp` paraleliza el loop de pixeles con `#pragma omp parallel for`. `omp_pt_sched.cpp` usa `schedule(runtime)` y selecciona `static`, `dynamic` o `guided` con `omp_set_schedule`, lo que permite barrer chunksize sin recompilar. El espacio 2D de pixeles se aplano a un indice `idx`; asi el chunksize representa pixeles y no filas completas. Cada iteracion escribe una posicion unica de `pixels[idx]` y usa un RNG local sembrado con `idx + 1`, por lo que no hay condiciones de carrera.

`numba_pt.py` implementa la misma logica con `@njit(parallel=True)` y `prange`. Numba compila las funciones a codigo nativo con LLVM durante la primera llamada; por eso el script hace un warm-up minimo antes de medir. A diferencia de OpenMP, donde el paralelismo se expresa mediante directivas al compilador C++, Numba traduce una subparte compatible de Python/NumPy y administra sus propios threads. El RNG de Numba es un LCG local por pixel para evitar estado global compartido.

Computador 1

Procesador: Intel(R) Core(TM) i7-9750H CPU @ 2.60GHz

Cores fisicos reportados: 6

Threads logicos reportados: 12

Threads por core: 2

Resultados seriales

Las mediciones usan wall-clock time. Si se piden varias repeticiones con `--reps`, el informe conserva el minimo, siguiendo la recomendacion de clase para reducir ruido experimental.

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
serial	serial_rt	scene.txt	small	1	serial	0	0.083447
serial	serial_pt	scene.txt	small	1	serial	0	4.075030
serial	serial_rt	scene.txt	medium	1	serial	0	1.378930
serial	serial_pt	scene.txt	medium	1	serial	0	64.689800
serial	serial_rt	scene.txt	large	1	serial	0	5.657600
serial	serial_pt	scene.txt	large	1	serial	0	533.884000
serial	serial_rt	scene_many.txt	small	1	serial	0	0.220340
serial	serial_pt	scene_many.txt	small	1	serial	0	6.176730
serial	serial_rt	scene_many.txt	medium	1	serial	0	3.484430
serial	serial_pt	scene_many.txt	medium	1	serial	0	98.009700
serial	serial_rt	scene_many.txt	large	1	serial	0	13.774700
serial	serial_pt	scene_many.txt	large	1	serial	0	782.014000

Barrido de scheduling OpenMP

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
schedule	omp_pt_sched	scene.txt	medium	1	static	1	64.443800
schedule	omp_pt_sched	scene.txt	medium	1	static	8	65.133800
schedule	omp_pt_sched	scene.txt	medium	1	static	32	64.336100
schedule	omp_pt_sched	scene.txt	medium	1	static	128	64.795100
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	1	64.453600
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	8	64.673600
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	32	64.586500
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	128	64.583800
schedule	omp_pt_sched	scene.txt	medium	1	guided	1	64.648200
schedule	omp_pt_sched	scene.txt	medium	1	guided	8	64.460700

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
schedule	omp_pt_sched	schedtxt	medium	1	guided	32	64.414900
schedule	omp_pt_sched	schedtxt	medium	2	static	1	38.203700
schedule	omp_pt_sched	schedtxt	medium	2	static	8	38.156000
schedule	omp_pt_sched	schedtxt	medium	2	static	32	38.031300
schedule	omp_pt_sched	schedtxt	medium	2	static	128	37.833600
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	1	37.810500
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	8	37.481700
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	32	37.379500
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	128	37.637400
schedule	omp_pt_sched	schedtxt	medium	2	guided	1	37.459300
schedule	omp_pt_sched	schedtxt	medium	2	guided	8	37.322800
schedule	omp_pt_sched	schedtxt	medium	2	guided	32	37.382200
schedule	omp_pt_sched	schedtxt	medium	4	static	1	21.527300
schedule	omp_pt_sched	schedtxt	medium	4	static	8	22.044500
schedule	omp_pt_sched	schedtxt	medium	4	static	32	22.337600
schedule	omp_pt_sched	schedtxt	medium	4	static	128	22.103300
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	1	21.980000
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	8	22.071200
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	32	22.139400
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	128	22.119800
schedule	omp_pt_sched	schedtxt	medium	4	guided	1	22.068800
schedule	omp_pt_sched	schedtxt	medium	4	guided	8	22.064400
schedule	omp_pt_sched	schedtxt	medium	4	guided	32	22.097700
schedule	omp_pt_sched	schedtxt	medium	8	static	1	14.257600
schedule	omp_pt_sched	schedtxt	medium	8	static	8	14.280900
schedule	omp_pt_sched	schedtxt	medium	8	static	32	14.330900
schedule	omp_pt_sched	schedtxt	medium	8	static	128	14.337100
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	1	14.067100
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	8	14.121400
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	32	14.137700
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	128	14.129000
schedule	omp_pt_sched	schedtxt	medium	8	guided	1	14.206500
schedule	omp_pt_sched	schedtxt	medium	8	guided	8	14.195200
schedule	omp_pt_sched	schedtxt	medium	8	guided	32	14.156600
schedule	omp_pt_sched	schedtxt	medium	1	static	1	99.549200
schedule	omp_pt_sched	schedtxt	medium	1	static	8	99.342300
schedule	omp_pt_sched	schedtxt	medium	1	static	32	99.531600
schedule	omp_pt_sched	schedtxt	medium	1	static	128	99.874400
schedule	omp_pt_sched	schedtxt	medium	1	dynamic	1	99.214900
schedule	omp_pt_sched	schedtxt	medium	1	dynamic	8	99.397000
schedule	omp_pt_sched	schedtxt	medium	1	dynamic	32	100.454000
schedule	omp_pt_sched	schedtxt	medium	1	dynamic	128	99.199800
schedule	omp_pt_sched	schedtxt	medium	1	guided	1	100.031000

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
schedule	omp_pt_sched_mandelbrot	medium		1	guided	8	99.319900
schedule	omp_pt_sched_mandelbrot	medium		1	guided	32	99.332400
schedule	omp_pt_sched_mandelbrot	medium		2	static	1	59.445000
schedule	omp_pt_sched_mandelbrot	medium		2	static	8	58.518200
schedule	omp_pt_sched_mandelbrot	medium		2	static	32	58.719800
schedule	omp_pt_sched_mandelbrot	medium		2	static	128	58.971500
schedule	omp_pt_sched_mandelbrot	medium		2	dynamic	1	58.627200
schedule	omp_pt_sched_mandelbrot	medium		2	dynamic	8	58.187600
schedule	omp_pt_sched_mandelbrot	medium		2	dynamic	32	58.370500
schedule	omp_pt_sched_mandelbrot	medium		2	dynamic	128	58.240100
schedule	omp_pt_sched_mandelbrot	medium		2	guided	1	58.552300
schedule	omp_pt_sched_mandelbrot	medium		2	guided	8	59.205600
schedule	omp_pt_sched_mandelbrot	medium		2	guided	32	57.949200
schedule	omp_pt_sched_mandelbrot	medium		4	static	1	34.195400
schedule	omp_pt_sched_mandelbrot	medium		4	static	8	35.180600
schedule	omp_pt_sched_mandelbrot	medium		4	static	32	35.624200
schedule	omp_pt_sched_mandelbrot	medium		4	static	128	35.205100
schedule	omp_pt_sched_mandelbrot	medium		4	dynamic	1	35.051100
schedule	omp_pt_sched_mandelbrot	medium		4	dynamic	8	35.141900
schedule	omp_pt_sched_mandelbrot	medium		4	dynamic	32	34.933700
schedule	omp_pt_sched_mandelbrot	medium		4	dynamic	128	34.955900
schedule	omp_pt_sched_mandelbrot	medium		4	guided	1	35.035800
schedule	omp_pt_sched_mandelbrot	medium		4	guided	8	34.922600
schedule	omp_pt_sched_mandelbrot	medium		4	guided	32	35.029800
schedule	omp_pt_sched_mandelbrot	medium		8	static	1	23.118700
schedule	omp_pt_sched_mandelbrot	medium		8	static	8	23.007300
schedule	omp_pt_sched_mandelbrot	medium		8	static	32	23.228800
schedule	omp_pt_sched_mandelbrot	medium		8	static	128	23.197100
schedule	omp_pt_sched_mandelbrot	medium		8	dynamic	1	22.729500
schedule	omp_pt_sched_mandelbrot	medium		8	dynamic	8	22.672800
schedule	omp_pt_sched_mandelbrot	medium		8	dynamic	32	22.885100
schedule	omp_pt_sched_mandelbrot	medium		8	dynamic	128	22.725000
schedule	omp_pt_sched_mandelbrot	medium		8	guided	1	23.687500
schedule	omp_pt_sched_mandelbrot	medium		8	guided	8	23.223600
schedule	omp_pt_sched_mandelbrot	medium		8	guided	32	23.011500

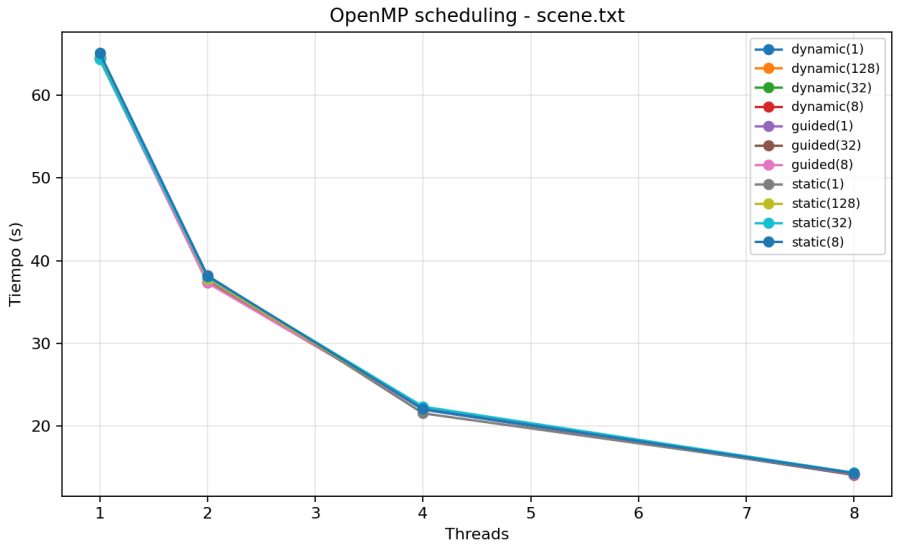
Comparacion OpenMP vs Numba

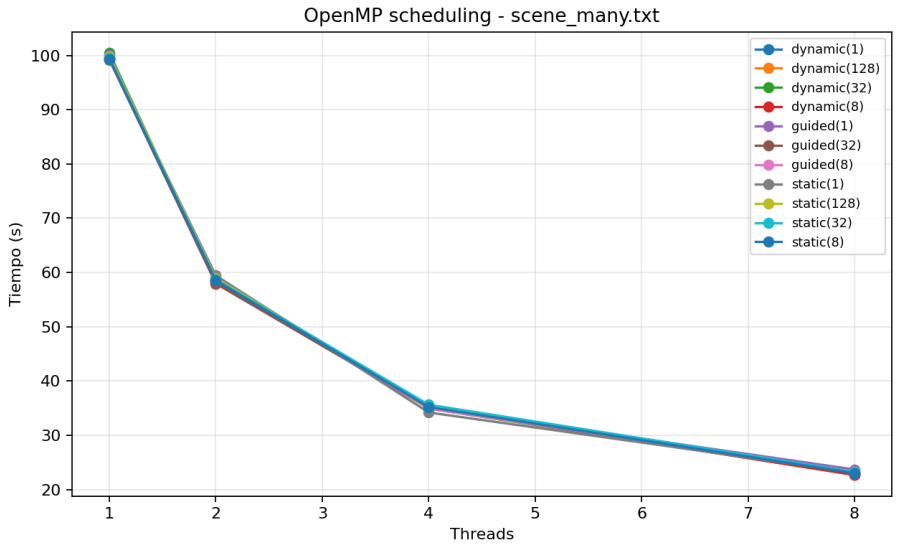
Para cada punto paralelo se calcula contra la referencia `serial_pt` de la misma escena y configuracion: `S(p) = Ts / T(p)` y `E(p) = S(p) / p`. Los graficos de tiempo incluyen ademas la curva ideal `Ts / p`.

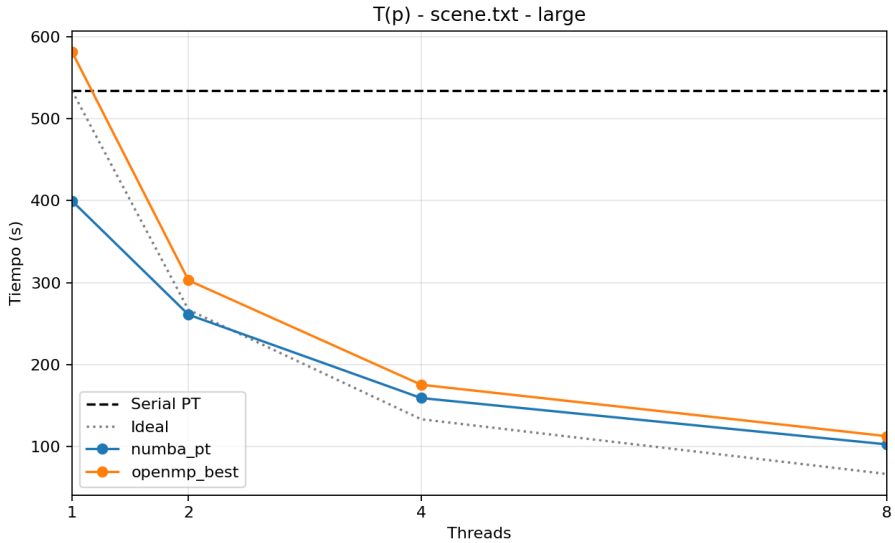
Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
compare	openmp_scene	test.txt	small	1	guided	8	4.181170
compare	openmp_scene	test.txt	small	2	guided	8	2.405550
compare	openmp_scene	test.txt	small	4	guided	8	1.346190
compare	openmp_scene	test.txt	small	8	guided	8	0.765462
compare	openmp_scene	test.txt	medium	1	guided	8	70.217400
compare	openmp_scene	test.txt	medium	2	guided	8	39.677700
compare	openmp_scene	test.txt	medium	4	guided	8	23.280200
compare	openmp_scene	test.txt	medium	8	guided	8	15.464000
compare	openmp_scene	test.txt	large	1	guided	8	581.278000
compare	openmp_scene	test.txt	large	2	guided	8	302.865000
compare	openmp_scene	test.txt	large	4	guided	8	175.561000
compare	openmp_scene	test.txt	large	8	guided	8	112.857000
compare	openmp_scene	test_many.txt	small	1	dynamic	128	6.307610
compare	openmp_scene	test_many.txt	small	2	dynamic	128	3.659360
compare	openmp_scene	test_many.txt	small	4	dynamic	128	2.227480
compare	openmp_scene	test_many.txt	small	8	dynamic	128	1.251090
compare	openmp_scene	test_many.txt	medium	1	dynamic	128	101.456000
compare	openmp_scene	test_many.txt	medium	2	dynamic	128	58.990300
compare	openmp_scene	test_many.txt	medium	4	dynamic	128	34.330200
compare	openmp_scene	test_many.txt	medium	8	dynamic	128	22.855100
compare	openmp_scene	test_many.txt	large	1	dynamic	128	796.495000
compare	openmp_scene	test_many.txt	large	2	dynamic	128	468.221000
compare	openmp_scene	test_many.txt	large	4	dynamic	128	279.796000
compare	openmp_scene	test_many.txt	large	8	dynamic	128	182.725000
compare	numba_pscene	test.txt	small	1	prange	0	3.578515
compare	numba_pscene	test.txt	small	2	prange	0	1.994008
compare	numba_pscene	test.txt	small	4	prange	0	1.142180
compare	numba_pscene	test.txt	small	8	prange	0	0.658976
compare	numba_pscene	test.txt	medium	1	prange	0	51.965004
compare	numba_pscene	test.txt	medium	2	prange	0	32.415745
compare	numba_pscene	test.txt	medium	4	prange	0	18.847139
compare	numba_pscene	test.txt	medium	8	prange	0	12.601827
compare	numba_pscene	test.txt	large	1	prange	0	399.574343
compare	numba_pscene	test.txt	large	2	prange	0	261.005850
compare	numba_pscene	test.txt	large	4	prange	0	159.329554
compare	numba_pscene	test.txt	large	8	prange	0	102.853527
compare	numba_pscene	test_many.txt	small	1	prange	0	5.487715
compare	numba_pscene	test_many.txt	small	2	prange	0	3.481544
compare	numba_pscene	test_many.txt	small	4	prange	0	2.048548
compare	numba_pscene	test_many.txt	small	8	prange	0	1.298125
compare	numba_pscene	test_many.txt	medium	1	prange	0	87.361184
compare	numba_pscene	test_many.txt	medium	2	prange	0	56.068161
compare	numba_pscene	test_many.txt	medium	4	prange	0	34.581465

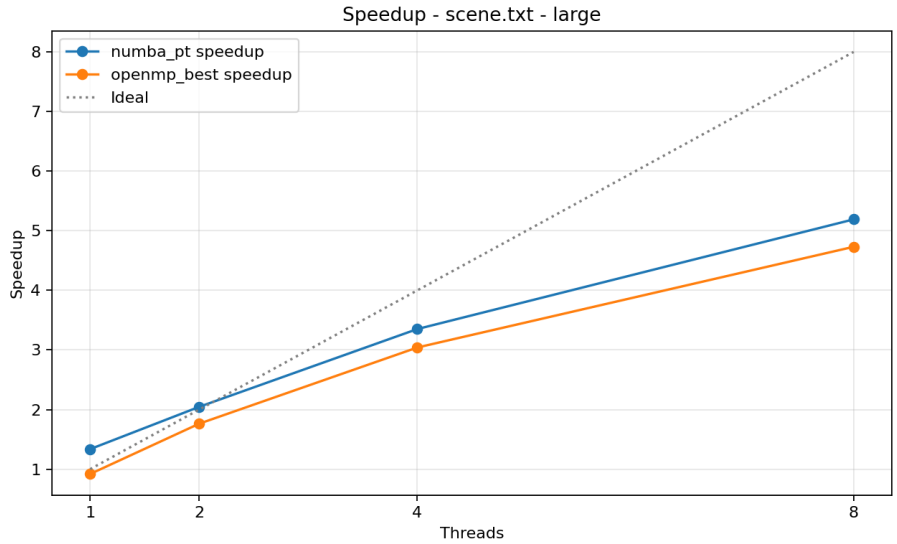
Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
compare	numba_p	scene_many.txt	medium	8	prange	0	23.217140
compare	numba_p	scene_many.txt	large	1	prange	0	696.564260
compare	numba_p	scene_many.txt	large	2	prange	0	452.235132
compare	numba_p	scene_many.txt	large	4	prange	0	284.197160
compare	numba_p	scene_many.txt	large	8	prange	0	185.490495

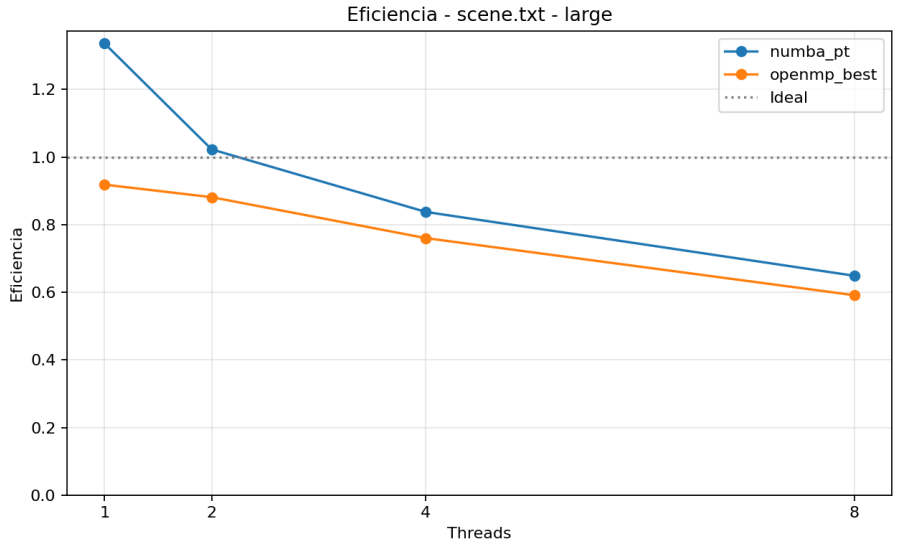
Graficos

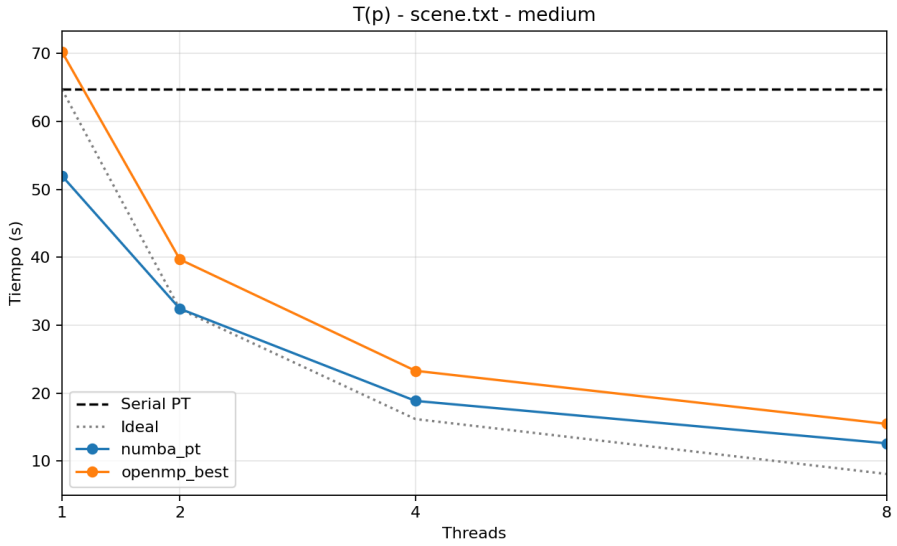

OpenMP scheduling - scene.txt
Tiempo (s)
Threads
dynamic(1)
dynamic(128)
dynamic(32)
dynamic(8)
guided(1)
guided(32)
guided(8)
static(1)
static(128)
static(32)
static(8)

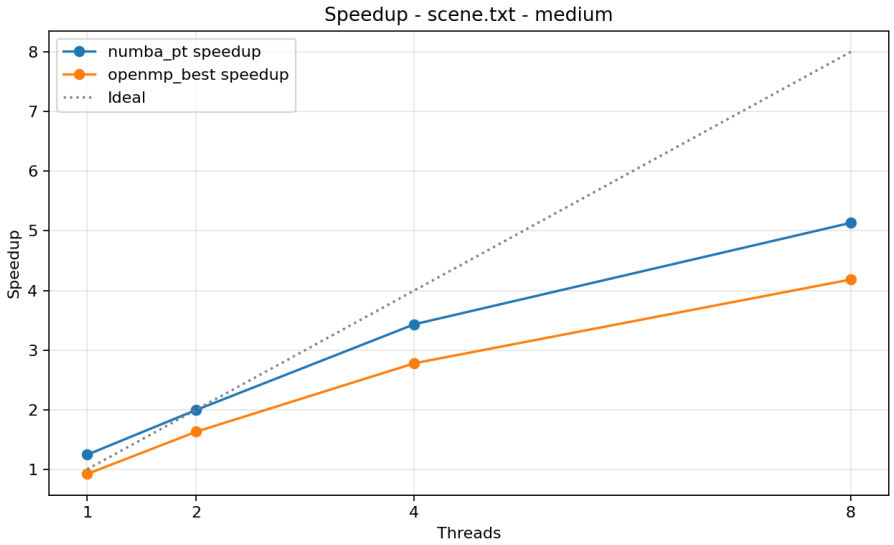

OpenMP scheduling - scene_many.txt
Tiempo (s)
Threads
dynamic(1)
dynamic(128)
dynamic(32)
dynamic(8)
guided(1)
guided(32)
guided(8)
static(1)
static(128)
static(32)
static(8)

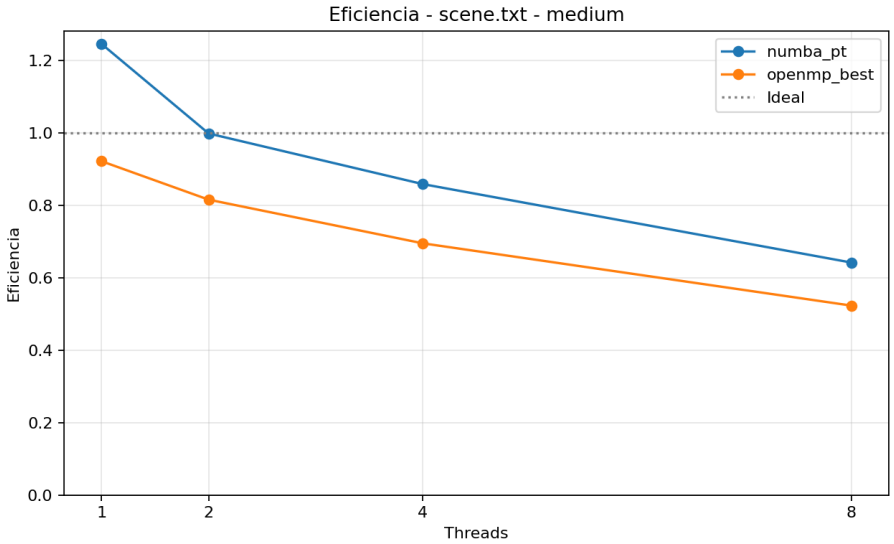

T(p) - scene.txt - large
Tiempo (s)
Threads
Serial PT
Ideal
numba_pt
openmp_best
600
500
400
300
200
100
1
2
4
8

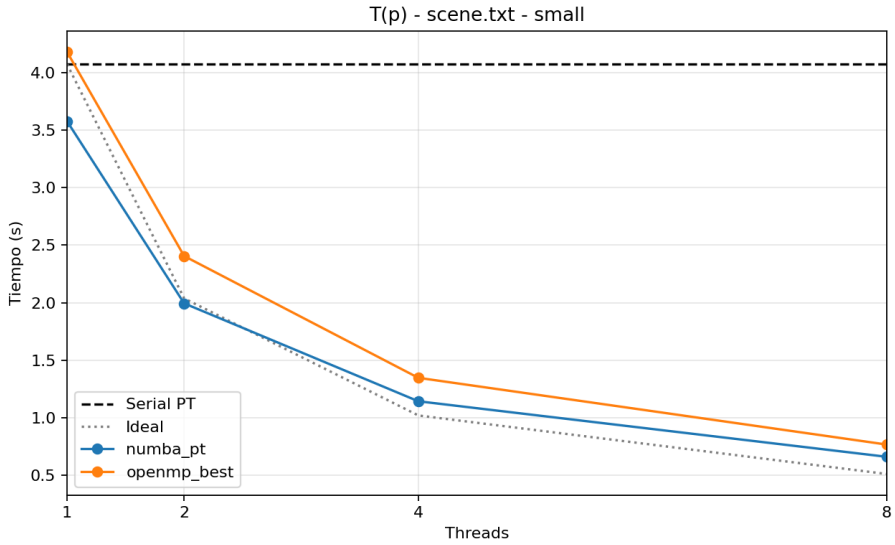

Speedup - scene.txt - large
Speedup
Threads
numba_pt speedup
openmp_best speedup
Ideal
8
7
6
5
4
3
2
1
1
2
4
8

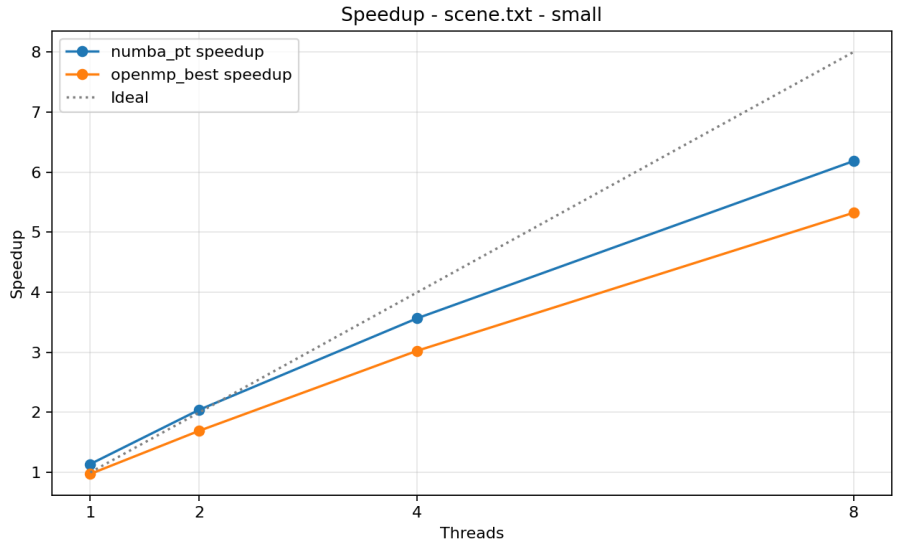

Eficiencia - scene.txt - large
numba_pt
openmp_best
Ideal
Eficiencia
Threads
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1
2
4
8

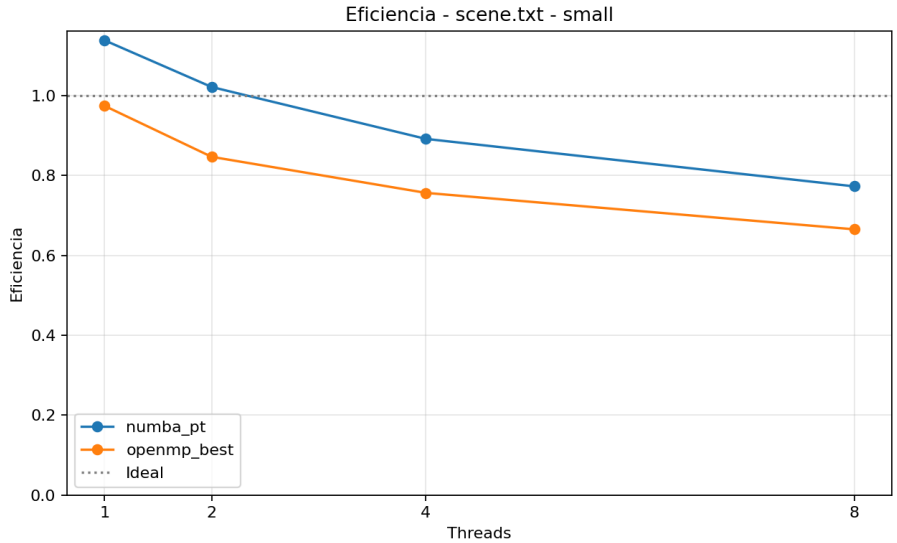

T(p) - scene.txt - medium
Serial PT
Ideal
numba_pt
openmp_best
Tiempo (s)
Threads
10
20
30
40
50
60
70
1
2
4
8

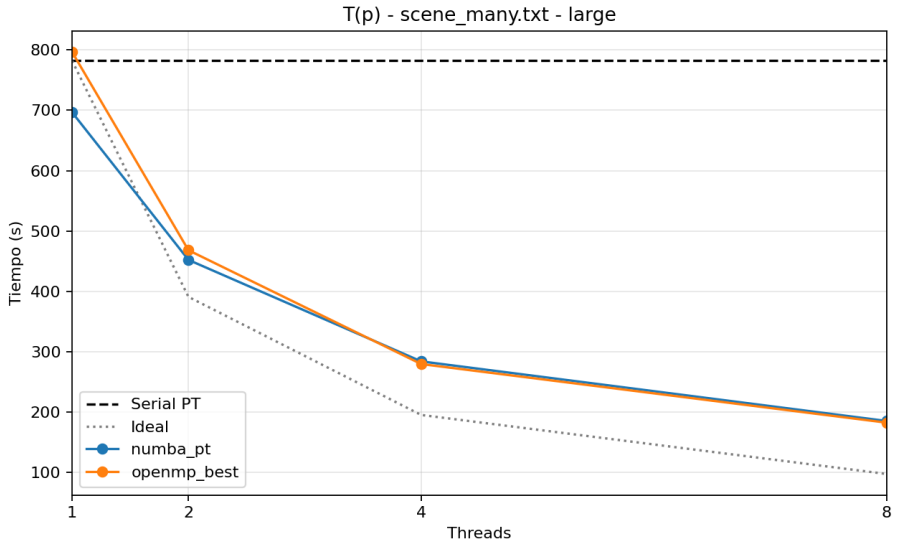

Speedup - scene.txt - medium
numba_pt speedup
openmp_best speedup
Ideal
Speedup
Threads
1
2
3
4
5
6
7
8
1
2
4
8

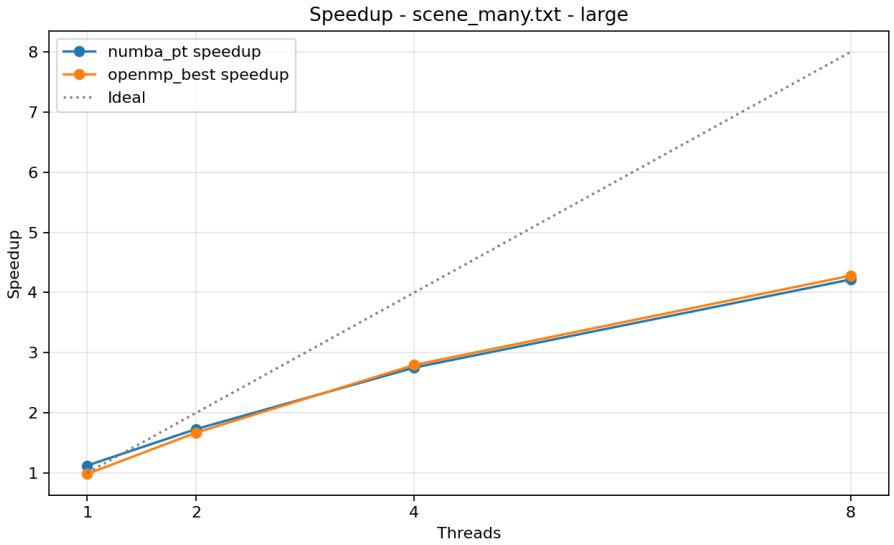

Eficiencia - scene.txt - medium
numba_pt
openmp_best
Ideal
Eficiencia
Threads
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1
2
4
8

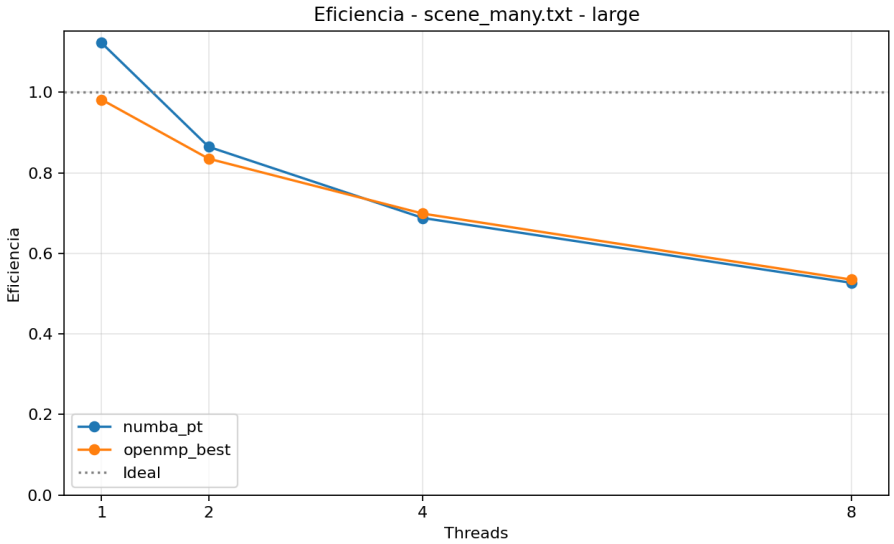

T(p) - scene.txt - small
Tiempo (s)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
Serial PT
Ideal
numba_pt
openmp_best
1
2
4
8
Threads

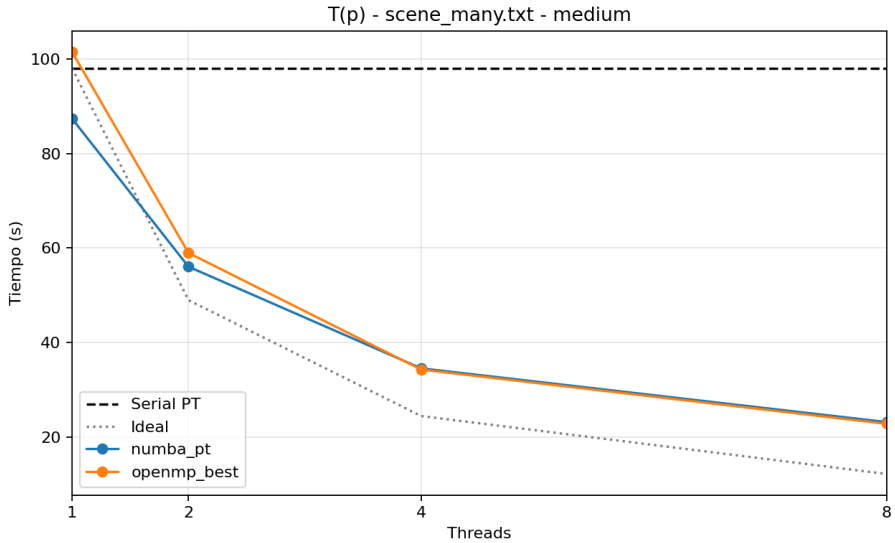

Speedup - scene.txt - small
Speedup
8
7
6
5
4
3
2
1
numba_pt speedup
openmp_best speedup
Ideal
1
2
4
8
Threads

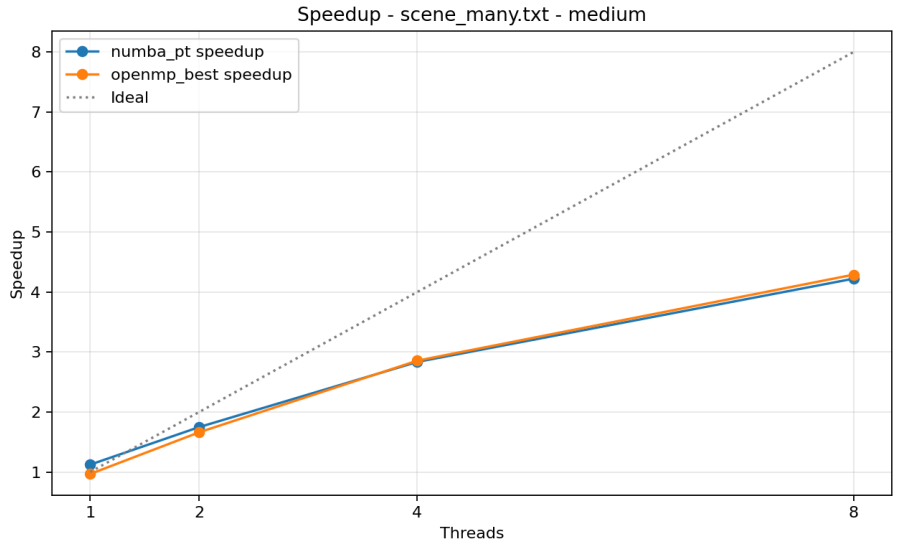

Eficiencia - scene.txt - small
Eficiencia
Threads
numba_pt
openmp_best
Ideal
0.0
0.2
0.4
0.6
0.8
1.0
1
2
4
8

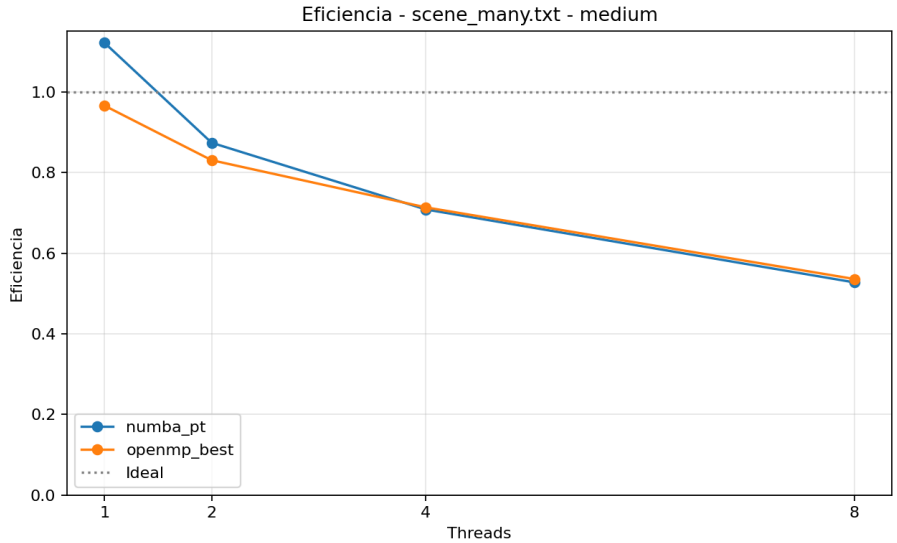

T(p) - scene_many.txt - large
Tiempo (s)
Threads
Serial PT
Ideal
numba_pt
openmp_best
100
200
300
400
500
600
700
800
1
2
4
8

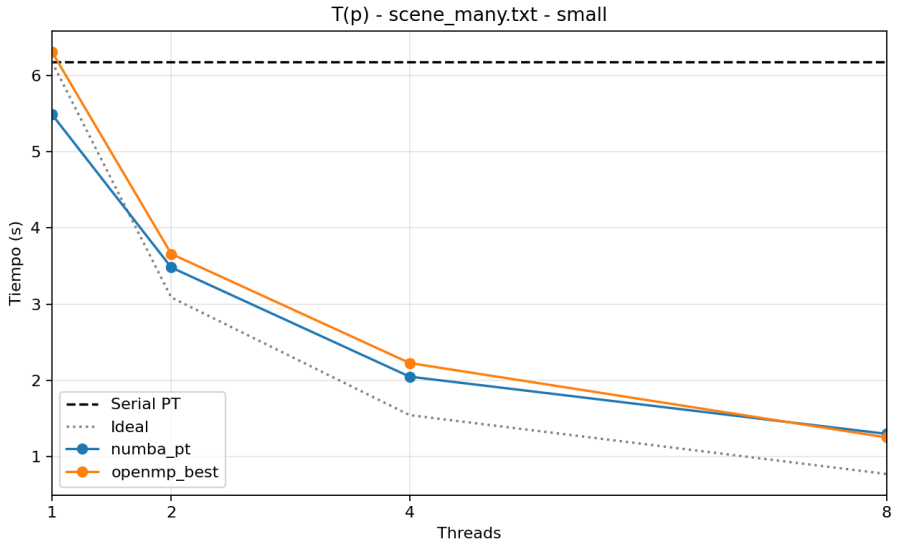

Speedup - scene_many.txt - large
numba_pt speedup
openmp_best speedup
Ideal
Speedup
Threads
1
2
4
8
1
2
3
4
5
6
7
8

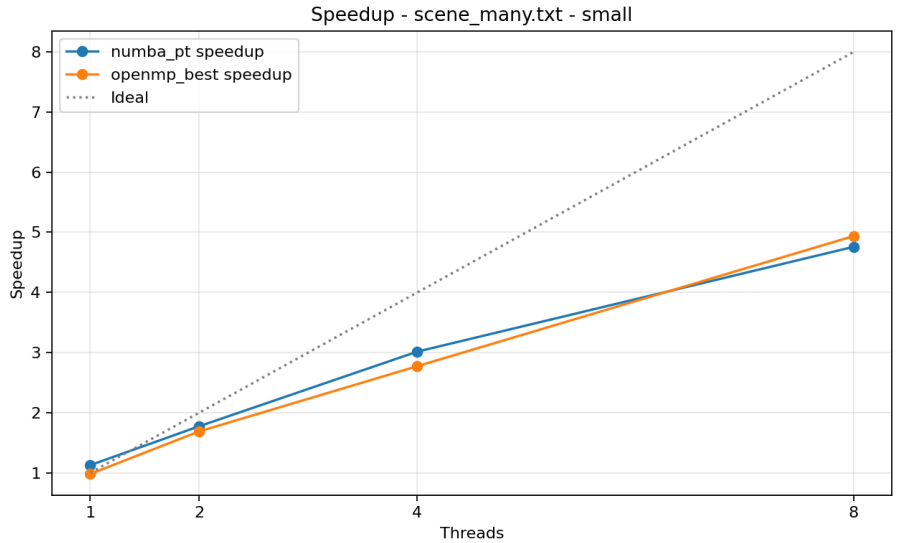

Eficiencia - scene_many.txt - large
numba_pt
openmp_best
Ideal
Eficiencia
Threads
1
2
4
8
0.0
0.2
0.4
0.6
0.8
1.0

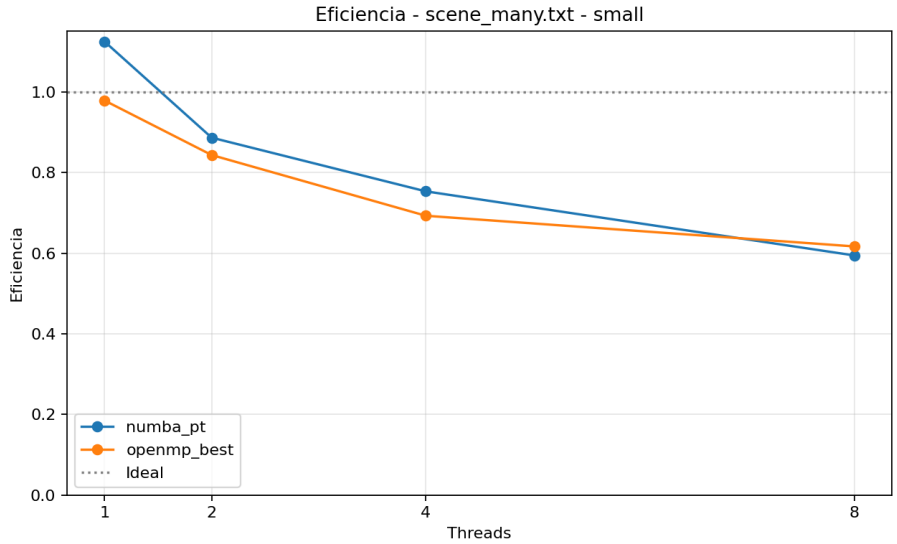

T(p) - scene_many.txt - medium
Tiempo (s)
Threads
Serial PT
Ideal
numba_pt
openmp_best
100
80
60
40
20
1
2
4
8


Speedup - scene_many.txt - medium
Speedup
Threads
numba_pt speedup
openmp_best speedup
Ideal
8
7
6
5
4
3
2
1
1
2
4
8


Eficiencia - scene_many.txt - medium
Eficiencia
Threads
numba_pt
openmp_best
Ideal
0.0
0.2
0.4
0.6
0.8
1.0
1
2
4
8


T(p) - scene_many.txt - small
Tiempo (s)
Threads
Serial PT
Ideal
numba_pt
openmp_best
1
2
3
4
5
6
1
2
4
8





Comentarios

El crecimiento esperado al duplicar la resolucion es cercano a 4x, porque se cuadruplica el numero de pixeles. En path tracing, duplicar N duplica aproximadamente el trabajo por pixel, salvo variaciones por terminacion anticipada y efectos de cache/runtime. La escena con 40 esferas aumenta K y por tanto el costo de cada interseccion, lo que vuelve mas visible el desbalance de carga.

La eficiencia decae al aumentar p por overhead fork-join, barreras implicitas,

planificacion dinamica, contencion de recursos compartidos y saturacion de partes del procesador. En esta tarea no hay reducciones ni escrituras compartidas sobre una misma variable; el principal riesgo de rendimiento es el desbalance y el costo de scheduling con chunks demasiado finos.

Segundo computador

Ejecute el mismo comando completo en otro computador y agregue aqui la tabla y graficos generados. Deben declararse cores fisicos, threads logicos y frecuencia del procesador de ese equipo.